

Arunabha Majumder

Mechatronics & Robotics Engineer | Mechanical Design | Actuators & Control | Prototyping

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Portfolio: arunabha22.github.io/arunabha-majumder | Project: www.viexo.aau.dk

PROFESSIONAL SUMMARY

Mechatronics and robotics engineer with a BTech in Mechanical Engineering, an MTech in Mechatronics and a PhD in progress at Aalborg University, with more than five years of hands-on R&D in robotic and electromechanical systems. Takes ideas from concept to tested hardware: mechanical design in SolidWorks, FEA, rapid prototyping, actuator and sensor integration, control implementation, and structured testing and validation. Designed a hybrid-actuated shoulder exoskeleton that cut energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026). Collaborative, structured and fluent in English (C1), with experience in international teams in Denmark and India.

CORE SKILLS

- **Mechanical Design:** Mechanical design, mechanisms, actuator design, 3D CAD, technical drawings, FEA, Design for Assembly (DFA)
- **Robotics:** Robotic systems, parallel robots, compliant and variable-stiffness actuation, exoskeletons, kinematics and Jacobian-based force estimation
- **Mechatronics & Integration:** Motors, encoders, force/torque sensors, pressure sensors, DAQ, embedded control (Arduino), pneumatics
- **Control & Modelling:** PID, neural-network gain scheduling, Assist-as-Needed control, MATLAB/Simulink, Python
- **Prototyping & Testing:** Rapid prototyping, machining, 3D printing, assembly, custom test rigs, test planning, data analysis
- **Collaboration:** Cross-disciplinary collaboration, technical writing, teaching and supervision

PROFESSIONAL EXPERIENCE

Research Engineer – Robotics & Actuator Development (PhD Project: VIEOXO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Designed a hybrid-actuated shoulder exoskeleton combining a parallel spring with a 6 Nm motor to offload gravity torque, cutting energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Built the full prototype from concept to tested hardware: SolidWorks CAD and FEA, machined and 3D-printed parts, assembly, motor, encoder, force-sensor and DAQ integration, embedded control and custom automated test rigs.
- Developed end-effector force estimation for a 5-bar planar parallel robot using variable-stiffness actuators (VSA) and the robot Jacobian, validated against a force sensor to below 10% RMS error.
- Implemented an Assist-as-Needed hybrid controller on the prototype, reducing user muscle effort by more than 15% in tests with human participants.
- Planned and ran a 12-participant validation study across 3 assistance modes and 2 load conditions, including statistical power justification, EMG and motion-capture instrumentation, ethics protocol and data analysis.
- Taught university students and supervised Bachelor's (B.Tech) students (2023 – 2025).

MTech Researcher – Pneumatic Actuator Systems (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Reduced position-tracking error to 0.3–0.78% with a neural-network-based gain-scheduled PID controller for a non-linear pneumatic actuator, outperforming classical PID under varying load.
- Fabricated pneumatic artificial muscles in-house from raw materials and built the test set-up with valves, pressure sensors and DAQ to characterise load–displacement behaviour and derive an empirical model.

Mechanical Engineering Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019

Siddaganga Institute of Technology, Tumakuru, India

- Led CAD design, fabrication and user testing end to end; won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant.

EDUCATION

PhD Candidate, Mechatronics and Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

MTech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

BTech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

PUBLICATIONS

- Majumder, A., Wagner, J.W., Zhu, Y., Oliveira, A.S., and Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." *Robotics and Mechatronics: ISRM 2026, MMS*, vol. 213, Springer.
- Majumder, A., Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., and Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." *IEEE CONECCT*, 2022.

CERTIFICATIONS & AWARDS

- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- Certified SolidWorks Associate (CSWA) – Mechanical Design
- Best Major Project Award, Department of Mechanical Engineering (2019)
- TATA Technologies Innovation Award – INR 100,000 Grant (2019)

HANDS-ON PROJECTS

- **Go-Kart (2017):** Led end-to-end design and fabrication of a custom Go-Kart.
- **Efficycle (2018):** Worked in a multidisciplinary team to build a human-electric hybrid vehicle for the Efficycle competition.

TECHNICAL TOOLS

- **CAD & Manufacturing:** SolidWorks, Autodesk Fusion 360, FEA, Bambu Lab and Ultimaker Cura (3D printing)
- **Software & Control:** MATLAB/Simulink, Python, LabVIEW (basic), Arduino, LaTeX
- **Instrumentation & Testing:** Qualisys Motion Capture, Forsentec Force/Torque Sensors, EMG, Pneumatic Control, Custom Rig Fabrication
- **Visual Communication:** Adobe Illustrator, Adobe Photoshop; Diploma in Fine Arts (Painting), Bangiya Sangeet Parishad

LANGUAGES

English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

REFERENCES

Available upon request.